

Research focus: Mathematical statistics and the theoretical foundations of causal inference/discovery, combined with extreme value theory, uncertainty quantification and time series.

CURRENT POSITION

- **Stanford University, Department of Statistics** Sep 2026 – Present
Stein Fellow Stanford, USA

EDUCATION

- **University of Lausanne (DO-HEC), Switzerland** Sep 2021 – July 2026
PhD candidate under Prof. Valérie Chavez-Demoulin Lausanne, Switzerland
 - PhD thesis: Estimating Unidentifiable Objects in Causal Inference
- **Charles University, Faculty of Mathematics and Physics** 2016 – 2021
BSc & MSc in Pure Mathematics and Mathematical Statistics Prague, Czech Republic
 - Excellence scholarship (grades in top few %)
 - Best Master's Thesis in Statistics (10,000 CZK award) under Prof. Zbyněk Pawlas
 - Research assistant on the Primus GeMS project for outstanding research

EXPERIENCE

- **University of Osaka, Faculty of Data Science** Dec 2025 – Feb 2026
Visiting Student Researcher Osaka, Japan
 - Collaborated with Prof. Shohei Shimizu on causal discovery.
- **UC Berkeley, Department of Statistics** Aug 2024 – Jan 2025
Visiting Student Researcher Berkeley, USA
 - Collaborated with Prof. Bin Yu on causal inference and uncertainty quantification.
- **Czech Academy of Sciences** Jun 2020 – Jun 2021
Research Assistant (part-time) Prague, Czech Republic
 - Conducted research under Prof. Milan Paluš; forming the basis of my master's thesis.

PUBLICATIONS

1. **J. Bodik** (2026). *Counterfactual prediction with cross-world dependence*. **CleaR**.
2. **J. Bodik**, V. Chavez-Demoulin (2025). *Identifiability of causal graphs under nonadditive conditionally parametric causal models*. **Journal of Machine Learning Research**.
3. **J. Bodik**, V. Chavez-Demoulin (2025). *Structural restriction in local causal discovery: identifying direct causes of a target variable*. **Biometrika**.
4. I. Azizi*, **J. Bodik***, J. Heiss*, B. Yu (2025). *CLEAR: calibrated learning for epistemic and aleatoric risks*. **ICLR**. *Co-first authors.
5. **J. Bodik** (2024). *Extreme treatment effect: extrapolating causal effects into extreme treatment domain*. **Mathematics**.
6. **J. Bodik**, Z. Pawlas, M. Paluš (2024). *Causality in extremes of time series*. **Extremes**.

PREPRINTS

1. **J. Bodik**, Y. Huang, B. Yu (2025). *Cross-world assumption and prediction intervals for individual treatment effects*. Preprint submitted to **JRSS B**. [arXiv:2507.12581](https://arxiv.org/abs/2507.12581)
2. **J. Bodik**, O. Pasche (2024). *Granger causality in extremes*. Preprint submitted to **Bernoulli**. [arXiv:2407.09632](https://arxiv.org/abs/2407.09632)
3. **J. Bodik**, L. Mhalla, V. Chavez-Demoulin (2023). *Detecting causal covariates for extreme dependence structures*. Unpublished semester project; [arXiv:2212.09831](https://arxiv.org/abs/2212.09831)

GRANTS

- **Travel Grant FUnil ($\approx 5,000$ USD)** *Dec 2025 – Feb 2026*
Travel grant supporting research visit to Osaka University
 - Visiting Student Researcher in Prof. Shohei Shimizu's group.
- **Hasler Foundation Research Grant ($\approx 10,000$ USD)** *Aug 2024 – Jan 2025*
Highly competitive grant supporting research visit to UC Berkeley
 - Visiting Student Researcher in Prof. Bin Yu's group.

HONORS & AWARDS

- **Best Master's Thesis Award (10,000 CZK)** 2021
Top three master's theses in probability, statistics, and econometrics at Charles University
- **Research Assistantship – Primus GeMS Project ($\approx 50,000$ CZK)** 2020–2021
Awarded a research assistantship on the Primus GeMS project for outstanding research.
- **Excellence Scholarship at Charles University** 2021
Awarded for grades in the top few %.
- **Bronze Medal – International Mathematics Competition (IMC)** 2020
Bulgaria
- **Winner (top 10) – National Mathematical Olympiad** 2016
Ranked among the national top 10 in the high-school mathematical Olympiad in Slovakia.

TEACHING & SERVICE

- **Course Instructor** *Stanford University*
 - **STATS 203** (Autumn 2026): Regression modeling and statistical inference.
- **Lead Teaching Assistant** *University of Lausanne*
 - **Optimization Methods** (2021–2023): Tutorials (lecture-type exercise sessions), exam preparation and grading, mentoring.
 - **Forecasting I** (2021–2025): Course coordination, grading, and advanced project mentoring.
 - **Forecasting II** (2021–2026): Tutorials, project supervision and consultations for MSc students.
 - **Risk Analytics** (2023–2025): Tutorials, exam preparation, grading, development and supervision of applied case studies for dozens of projects.
- **Trojsten NGO - Volunteer Educator and Event Organizer** *Bratislava, Slovakia, 2016–2022*
 - Delivered 80+ math lectures to high-school students, trained national math Olympiad winners, and organized math competitions with 200+ participants.
- **Peer Review**
 - Annals of Statistics, JRSS B; JASA, Scandinavian journal of statistics, ICLR, Statistics and Probability Letters; Behaviormetrika; REVSTAT; Data Mining and Knowledge Discovery
- **First Committee Member**
Expertise Center for Climate Extremes (ECCE, UNIL)
 - First member of a PhD committee at a newly established research center; responsible for outreach and seminar & event organization.
- **Seminar Coordination**
 - (**Stanford University, Statistics Seminars**) Primary coordinator: managed speaker outreach, budgeting and travel, publicity, and post-talk meetings.
 - (**University of Lausanne, HEC–DO Seminars**) Primary coordinator: managed speaker outreach, budgeting and travel, publicity, and post-talk meetings.

INVITED TALKS

- **OCIS (Online Causal Inference Seminar)** **Online Feb 2026**
Invited speaker at the online causal inference seminar (the largest seminar in the field).
- **Data science seminar** **Osaka, Japan Dec 2025**
Invited speaker at SANKEN+RIKEN - Data science and AI lab at University of Osaka.
- **CHUV BDSC Seminar** **Lausanne, Switzerland Sep 2025**
Invited speaker at the Clinical Data Science Group, CHUV University Hospital.
- **UAI 2024 Workshop on Causal Inference in Time Series** **Barcelona, Spain Jul 2024**
Invited speaker; one of seven invited speakers including Miguel Hernán and Gustau Camps-Valls.

CONTRIBUTED TALKS

Shimizu lab seminar <i>Speaker at Shimizu lab seminar at SANKEN, University of Osaka.</i>	Osaka, Japan Dec 2025
Data Driven Decisions and Inference <i>Speaker at a weekly internal research seminar hosted by Lihua Lei, Stefan Wager, and Jann Spiess.</i>	Stanford, USA Nov 2024
Bin Yu Group Seminar, Berkeley's Casual Causal Seminar and PhD Students Seminar <i>Presentations on methodological aspects of causal discovery and extreme value theory.</i>	Berkeley, USA Oct 2024
Causal Science Center Conference <i>Contributed talk on causality and extremes.</i>	Stanford, USA Oct 2024
CMStatistics 2023 <i>Speaker on causal discovery.</i>	Berlin, Germany Dec 2023
EuroCIM: European Causal Inference Meeting <i>Speaker on causal discovery.</i>	Oslo, Norway Apr 2023
BIRS: Combining Causal Inference and Extreme Value Theory in the Study of Climate Extremes <i>Speaker at a Banff International Research Station workshop</i>	Kelowna, Canada (online) Jun 2022
Robust Conference <i>Speaker: presentation on causality in extremes of time series.</i>	Prague, Czech Republic Jun 2022
CMStatistics 2021 <i>Speaker: presentation on causality in extremes of time series.</i>	London, UK (online) Dec 2021
Valpred Workshop <i>Speaker: presentation on causality in extremes of time series.</i>	Aussois, France Oct 2021
Extreme Value Analysis Conference <i>Speaker: Presentation on causality in extremes of time series.</i>	Edinburgh, UK (online) Jun 2021

POSTERS AT CONFERENCES/WORKSHOPS

CLear conference at Broad Institute of MIT and Harvard <i>Poster on counterfactual inference</i>	Cambridge, USA April 2026
Causal identification and discovery workshop at Newton Institute <i>Poster on causal discovery</i>	Cambridge, UK March 2026
Causal Science Center Conference <i>Poster on a conference hosted by Stanford Causal Science Center.</i>	Stanford, USA Oct 2024
EGU24 <i>Poster on causal inference and extremes presented in the section "Many shades of causality"</i>	Vienna, Austria Apr 2024
Causality in Extremes Workshop <i>Poster at the University of Geneva workshop on causality in extremes.</i>	Geneva, Switzerland Feb 2024

SKILLS

- **Programming:** R, Python, $\text{\LaTeX}$; primarily use R for academic work.
- **Languages:** English (fluent), Czech (fluent), French (upper intermediate), German (lower intermediate), Japanese (only basics), Slovak (native)

SELECTED CODE & REPOSITORIES

• ITE_prediction_cross_world <i>Causal ITE prediction under cross-world assumptions</i>	
• structural-restrictions-in-local-causal-discovery <i>Structural restrictions in local causal discovery</i>	
• Granger-causality-in-extremes <i>Causality in extremes of time series</i>	
• Extreme_treatment_effect <i>Extrapolating causal effects into extreme treatment domain</i>	
• causal_cpcm <i>Conditionally parametric causal models</i>	

OTHER ACTIVITIES

- **Ultimate Frisbee – World Championships**

2016-ongoing

National Representation

- Represented Slovakia at the U20 World Championships
- Also competed for the Lausanne club *Flyhigh* at the Swiss National Championships

- **Cipher Competition Series**

2020

Main organizer of an event with 250+ participants.